Red Team Critique of Blue Team Version Four

Version four adopts better vocabulary and fixes several earlier points: it names oblivious pseudorandom functions, adds reversible actions, sets an appeal deadline, and drops the Kalman-free pretense of version three's thirty-day clearing rule. But its central claim, that it directly addresses every critical fault, is false. The headline correction is not a correction, two features remain mathematically wrong, and several fixes from earlier rounds quietly disappeared.

Critical faults

1. The "corrected" capacity formula is identical to the one the Red Team rejected.
Version four again divides total alerts by target recall and then by precision. Using the figures from version three's own example, 12,857 alerts divided by 0.9 and then by 0.14 gives about 102,000 reviews, roughly eight times too many. The correct relationship is unchanged: reviews needed equal abusers multiplied by target recall, divided by precision, which is 2,000 times 0.9 divided by 0.14, or about 12,857. Further problems:

- The worked example has vanished, so version four contains formulas with no numbers. Targets for precision and recall are never stated, and thresholds are again the placeholder "X".
- The latency formula adds an analyst term to a legal term, but the share of cases sent to legal review is never defined, and the formula says nothing about waiting time. A review team running near full utilization has queues that grow sharply, so capacity must be sized with spare room.
- There is no overflow rule. When the legal queue is full, do held funds stay held, which harms honest merchants, or get released, which helps abusers? That default is a core policy decision.

2. The Flow Divergence Index cannot separate anything.
The numerator is outflow minus inflow. A collection account that sweeps money onward has outflow close to inflow, so the numerator is near zero, and multiplying a near-zero number by any diversity factor leaves it near zero. The same near-zero value describes a pass-through account, a business that spends everything it earns, and a perfectly balanced account. Other problems:

- "Total transaction value" is undefined.

- The index treats high payer diversity as a sign of legitimate commerce. But a betting operator's account also has many distinct payers. The discriminating quantity is how often diverse payers return, compared with their cohort.
- The summary claims the index captures dwell time. The formula contains no dwell time. The signal the second critique asked for, how long money stays in the account and what share of inflow leaves within a short window, is still missing.

3. The closed-loop signature is the best idea in version four, but it is defined backwards and too strictly.
As written, payments made by a payer are followed by payments received from the same payer, which describes repeat payments in one direction. A closed loop means money going out to people who earlier paid in. Further problems:

- The first condition, frequent payments to distinct external entities, also matches wages, refunds, and cashback.
- "Correlating precisely" is unrealistic. Most bettors lose and receive nothing, so the loop covers only winners, and amounts follow odds, not exact stakes. The signal should be statistical: the share of outflow going to earlier payers, and the ratio of payout to prior stake.
- Operators often pay winnings from a separate account or in cash, which splits the loop across accounts the graph must first link.

4. The random audits would exceed capacity and still would not fix the label problem.
Suppose nine tenths of one million merchants are cleared. Auditing ten percent of them is 90,000 reviews, about seven times everything the detection plan needed. Yield is also tiny. If missed abusers are 0.02 percent of the cleared group, about 200 merchants, those 90,000 audits find about 20, and an affordable sample of 2,000 audits finds fewer than one on average. Better design samples more heavily from riskier strata while recording each merchant's inclusion probability, so unbiased estimates remain possible, and adds other label sources such as payer complaints and regulator referrals. The protected holdout set, the standard for what counts as a confirmed case, and the handling of slow investigations are still absent.

5. The Kalman filter has returned, and it still follows drift instead of detecting it.
A Kalman filter tracks a moving level. A merchant creeping toward illicit volume is tracked, and the tracked value becomes normal. What the second critique asked for was a cumulative change test measured against the fixed onboarding profile and the cohort. Version four's cohort trigger answers a different problem, many merchants drifting together, and does nothing for a single honest merchant who changes business and needs to re-declare a profile. Calling it a defense against tier gaming is a non sequitur, because tier gaming means bunching just under a threshold.

Major faults

6. Account control still points the wrong way.

- Version four says delegation is flagged when activity matches shared patterns among several parties. Delegation by family or employees was meant to be treated as normal and excluded from the rental signal. Flagging it penalizes exactly the honest shops. A shared pattern among several operators is also what a rented account looks like, so the rule is circular.
- Three logins from three continents in one day is a clear case of credential compromise, but rented accounts are usually operated from one consistent place, often abroad.
- Missing signals from the previous critique: login location compared with the shop's address and opening hours, whether the owner's own devices still appear, who initiates payouts, and how far payers live from the shop.
- A product-level fix is available: let merchants enroll named, verified authorized users and devices. Unlisted operators then stand out, and honest delegation becomes visible instead of suspicious.

7. The linkage description contradicts itself.

Oblivious pseudorandom functions are the right family of tools, but version four says matching identifiers are revealed while values remain "unlinkable across providers". Linkage across providers is the entire purpose. It still proposes hashed network ranges, which link unrelated merchants on shared mobile networks and do not need hashing in such a protocol anyway. Standard protocols serve two parties, and extending them to many providers plus a central system changes the cost and the guarantees. The protocol outputs matches, not risk scores, so a second step is needed. Finally, a protocol cannot define liability or who owns a key. That requires a legal basis, a contract or regulation, calibration across providers, and a correction process with real content.

8. The legal triage layer and the reversible measure contradict each other.

- A temporary settlement delay is a hold. Small merchants often depend on daily settlement to buy stock, so it cannot be described as keeping money flowing. It also occurs before legal triage, contradicting the rule that no restrictive action happens first. A more proportionate form holds only an amount matching the estimated suspicious flow, for a stated maximum number of days, under a contractual right.
- Counsel cannot define an action as a function of a zero-to-one-hundred score and jurisdiction. Counsel decides which evidence justifies which action. Statisticians set calibrated score bands and the precision expected in each band, and operations sets capacity. The score is never said to be calibrated, so no legal threshold can attach to it.

9. Due process leaks signals and overclaims.

A rationale code citing the specific feature violation, such as the index exceeding a threshold, tells an abuser which signal to avoid. Version three's two levels of explanation, a general one for the merchant and a detailed one for reviewers, have disappeared. Neutral wording does not "eliminate" tipping-off risk, because that depends on the law and on whether a report exists. The appeal deadline of seven days is a good addition, but version four says nothing about funds during the appeal, what happens when the deadline is missed, or whether the appeal reviewer is independent of the original analyst.

Moderate faults

10. Consistency testing contradicts threshold randomization. Randomized thresholds guarantee that similar merchants sometimes receive different treatment, and then the shadow test flags the system for bias. Two merchants never have identical risk profiles, so similarity needs a definition. Individual consistency is also not group fairness: the test should compare outcomes across merchant size, region, and formal versus informal businesses.

11. Several earlier faults remain unanswered while the summary claims otherwise.

- Insider and process risk has now gone unanswered for three rounds.
- The cost model has no numbers: no alert volume, reviewer hours, or graph computation cost.
- The event calendar, all-day games, payer concentration, payouts through cash, and thin digital footprints among informal merchants are not addressed.
- Backtesting and champion-challenger comparison, present in version two, have vanished.

What Blue Team version five should fix

Fault	Required fix
Capacity formula	Use reviews equal abusers times recall divided by precision, state target numbers, add throughput and queueing, and define the overflow rule
Flow index	Replace it with dwell time, the share of inflow leaving within a short window, and repeat rate among payers, all relative to the cohort
Closed loop	Measure the share of outflow going to earlier payers and payout-to-stake ratios, and link accounts so split loops are visible
Random audits	Sample by risk stratum with recorded inclusion probabilities, add a protected holdout set, define confirmed cases, and add other label sources
Drift	Use cumulative change tests against fixed anchors, and give each merchant a way to re-declare a profile
Account control	Enroll authorized users and devices, and add login location against shop address, payout initiator, and payer distance
Linkage	Specify the party setting, drop hashed network ranges, and write governance, legal basis, liability, and calibration
Holds	Hold only the suspicious portion, for a stated maximum time, and run counsel's evidence rules separately from calibrated score bands

Explanations	Restore two levels of explanation, state funds treatment during appeal, and add an independent reviewer
Missing items	Answer insider risk, scale cost, the event calendar, informal-economy bias, and backtesting

Bottom line: version four changed its tone more than its mathematics. The capacity formula, the central feature, and the draft's own summary all fail a simple check. The strongest next version would fix the arithmetic, define every feature precisely enough that someone could compute it, and include one end-to-end numerical example showing how many merchants flow through each stage.

I can draft Blue Team version five on that basis, or put this critique into a document if you would like.